MILITARY SPECIFICATION SHEET

DELAY LINES, 14 PIN DIP COMPATIBLE, 5 TAP
(10 SECTION MINIMUM)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the delay lines described herein shall consist of this specification and the latest issue of MIL-D-83532.

REQUIREMENTS:

Design and construction: Parts shall be of the design, construction and physical dimensions specified in the appendix of MIL-D-83532. Four case styles (A, B, C, and D) are available.

Input pulse: Delay lines must be capable of passing a minimum input pulse width of 25 percent of total delay time.

Schematic: See figure 1.

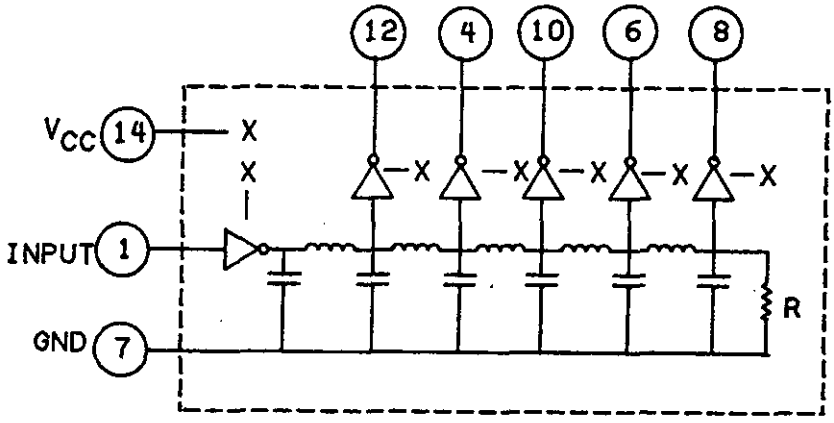
Delay times: Delay time from input to all taps shall be as specified in table I (+25°C) and table II (-55°C and +125°C).

DC parameters (over operating temperature range): See table III.

Rated maximum load (fan-out): Ten TTL Schottky loads per tap (no more than 20 TTL Schottky loads per unit).

Terminations: Part terminations snall be printed circuit pin type in accordance with MIL-M-38510.

Operating temperature: Operating temperature range shall be -55°C to +125°C.




NOTES:
1. Schematic is for general information only.
2. Pins 2, 3, 5, 9, 11 and 13 are omitted on Level A delay lines.
3. All pins are installed on Level B delay lines.

FIGURE 1. Schematic.


DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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TABLE I. Dash numbers and delay characteristics at +25°C, $V_{CC}$ = 5.00 ±.01 volts.

| Dash number | Delay and tolerances (ns) | | | | |
|---|---|---|---|---|---|
| | TAP 1 PIN 12 | TAP 2 PIN 4 | TAP 3 PIN 10 | TAP 4 PIN 6 | OUTPUT PIN 8 |
| -001 | 5.0 ±2 ns | 10 ±2 ns | 15 ±2 ns | 20 ±2 ns | 25 ±2 ns |
| -002 | 6.0 ±2 ns | 12 ±2 ns | 18 ±2 ns | 24 ±2 ns | 30 ±2 ns |
| -003 | 7.0 ±2 ns | 14 ±2 ns | 21 ±2 ns | 28 ±2 ns | 35 ±2 ns |
| -004 | 8.0 ±2 ns | 16 ±2 ns | 24 ±2 ns | 32 ±2 ns | 40 ±2 ns |
| -005 | 9.0 ±2 ns | 18 ±2 ns | 27 ±2 ns | 36 ±2 ns | 45 ±5% |
| -006 | 10. ±2 ns | 20 ±2 ns | 30 ±2 ns | 40 ±2 ns | 50 ±5% |
| -007 | 11. ±2 ns | 22 ±2 ns | 33 ±2 ns | 44 ±5% | 55 ±5% |
| -008 | 12. ±2 ns | 24 ±2 ns | 36 ±2 ns | 48 ±5% | 60 ±5% |
| -009 | 13. ±2 ns | 26 ±2 ns | 39 ±2 ns | 52 ±5% | 65 ±5% |
| -010 | 14. ±2 ns | 28 ±2 ns | 42 ±5% | 56 ±5% | 70 ±5% |
| -011 | 15. ±2 ns | 30 ±2 ns | 45 ±5% | 60 ±5% | 75 ±5% |
| -012 | 16. ±2 ns | 32 ±2 ns | 48 ±5% | 64 ±5% | 80 ±5% |
| -013 | 18. ±2 ns | 36 ±2 ns | 54 ±5% | 72 ±5% | 90 ±5% |
| -014 | 20. ±2 ns | 40 ±2 ns | 60 ±5% | 80 ±5% | 100 ±5% |
| -015 | 25. ±2 ns | 50 ±5% | 75 ±5% | 100 ±5% | 125 ±5% |
| -016 | 30. ±2 ns | 60 ±5% | 90 ±5% | 120 ±5% | 150 ±5% |
| -017 | 35. ±2 ns | 70 ±5% | 105 ±5% | 140 ±5% | 175 ±5% |
| -018 | 40. ±2 ns | 80 ±5% | 120 ±5% | 160 ±5% | 200 ±5% |
| -019 | 45. ±5% | 90 ±5% | 135 ±5% | 180 ±5% | 225 ±5% |
| -020 | 50. ±5% | 100 ±5% | 150 ±5% | 200 ±5% | 250 ±5% |
| -021 | 55. ±5% | 110 ±5% | 165 ±5% | 210 ±5% | 275 ±5% |
| -022 | 60. ±5% | 120 ±5% | 180 ±5% | 240 ±5% | 300 ±5% |
| -023 | 70. ±5% | 140 ±5% | 210 ±5% | 280 ±5% | 350 ±5% |
| -024 | 80. ±5% | 160 ±5% | 240 ±5% | 320 ±5% | 400 ±5% |
| -025 | 90. ±5% | 180 ±5% | 270 ±5% | 360 ±5% | 450 ±5% |
| -026 | 100 ±5% | 200 ±5% | 300 ±5% | 400 ±5% | 500 ±5% |
| -027 | 120 ±5% | 240 ±5% | 360 ±5% | 480 ±5% | 600 ±5% |
| -028 | 140 ±5% | 280 ±5% | 420 ±5% | 560 ±5% | 700 ±5% |
| -029 | 160 ±5% | 320 ±5% | 480 ±5% | 640 ±5% | 800 ±5% |
| -030 | 180 ±5% | 360 ±5% | 540 ±5% | 720 ±5% | 900 ±5% |
| -031 | 200 ±5% | 400 ±5% | 600 ±5% | 800 ±5% | 1000 ±5% |



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TABLE II. Dash numbers and delay characteristics at -55°C and +125°C, $V_{CC}$ = 5.00 ±.01 volts.

| Dash number | Delay and tolerances (ns) | | | | |
|---|---|---|---|---|---|
| | TAP 1 PIN 12 | TAP 2 PIN 4 | TAP 3 PIN 10 | TAP 4 PIN 6 | OUTPUT PIN 8 |
| -001 | 5.0 ±3 ns | 10 ±3 ns | 15 ±3 ns | 20 ±3 ns | 25 ±3 ns |
| -002 | 6.0 ±3 ns | 12 ±3 ns | 18 ±3 ns | 24 ±3 ns | 30 ±3 ns |
| -003 | 7.0 ±3 ns | 14 ±3 ns | 21 ±3 ns | 28 ±3 ns | 35 ±3 ns |
| -004 | 8.0 ±3 ns | 16 ±3 ns | 24 ±3 ns | 32 ±3 ns | 40 ±3 ns |
| -005 | 9.0 ±3 ns | 18 ±3 ns | 27 ±3 ns | 36 ±3 ns | 45 ±8% |
| -006 | 10. ±3 ns | 20 ±3 ns | 30 ±3 ns | 40 ±3 ns | 50 ±8% |
| -007 | 11. ±3 ns | 22 ±3 ns | 33 ±3 ns | 44 ±8% | 55 ±8% |
| -008 | 12. ±3 ns | 24 ±3 ns | 36 ±3 ns | 48 ±8% | 60 ±8% |
| -009 | 13. ±3 ns | 26 ±3 ns | 39 ±3 ns | 52 ±8% | 65 ±8% |
| -010 | 14. ±3 ns | 28 ±3 ns | 42 ±8% | 56 ±8% | 70 ±8% |
| -011 | 15. ±3 ns | 30 ±3 ns | 45 ±8% | 60 ±8% | 75 ±8% |
| -012 | 16. ±3 ns | 32 ±3 ns | 48 ±8% | 64 ±8% | 80 ±8% |
| -013 | 18. ±3 ns | 36 ±3 ns | 54 ±8% | 72 ±8% | 90 ±8% |
| -014 | 20. ±3 ns | 40 ±3 ns | 60 ±8% | 80 ±8% | 100 ±8% |
| -015 | 25. ±3 ns | 50 ±8% | 75 ±8% | 100 ±8% | 125 ±8% |
| -016 | 30. ±3 ns | 60 ±8% | 90 ±8% | 120 ±8% | 150 ±8% |
| -017 | 35. ±3 ns | 70 ±8% | 105 ±8% | 140 ±8% | 175 ±8% |
| -018 | 40. ±3 ns | 80 ±8% | 120 ±8% | 160 ±8% | 200 ±8% |
| -019 | 45. ±8% | 90 ±8% | 135 ±8% | 180 ±8% | 225 ±8% |
| -020 | 50. ±8% | 100 ±8% | 150 ±8% | 200 ±8% | 250 ±8% |
| -021 | 55. ±8% | 110 ±8% | 165 ±8% | 210 ±8% | 275 ±8% |
| -022 | 60. ±8% | 120 ±8% | 180 ±8% | 240 ±8% | 300 ±8% |
| -023 | 70. ±8% | 140 ±8% | 210 ±8% | 280 ±8% | 350 ±8% |
| -024 | 80. ±8% | 160 ±8% | 240 ±8% | 320 ±8% | 400 ±8% |
| -025 | 90. ±8% | 180 ±8% | 270 ±8% | 360 ±8% | 450 ±8% |
| -026 | 100 ±8% | 200 ±8% | 300 ±8% | 400 ±8% | 500 ±8% |
| -027 | 120 ±8% | 240 ±8% | 360 ±8% | 480 ±8% | 600 ±8% |
| -028 | 140 ±8% | 280 ±8% | 420 ±8% | 560 ±8% | 700 ±8% |
| -029 | 160 ±8% | 320 ±8% | 480 ±8% | 640 ±8% | 800 ±8% |
| -030 | 180 ±8% | 360 ±8% | 540 ±8% | 720 ±8% | 900 ±8% |
| -031 | 200 ±8% | 400 ±8% | 600 ±8% | 800 ±8% | 1000 ±8% |



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TABLE III. DC characteristics.

| Test | Symbol | Conditions<br>$-55°C \leq T_C \leq +125°C$ | Limits | | Unit |
|---|---|---|---|---|---|
| | | | Min | Max | |
| High-level output voltage | $V_{OH}$ | $V_{CC}$ = 4.5 V<br>$V_{IH}$ = 2.0 V<br>$I_{OH}$ = -1 mA | 2.5 | | V |
| Low-level output voltage | $V_{OL}$ | $V_{CC}$ = 4.5 V<br>$V_{IL}$ = 0.8 V<br>$I_{OL}$ = 20 mA | | 0.5 | V |
| Input clamp voltage | $V_{IC}$ | $V_{CC}$ = 4.5 V<br>$I_I$ = -18 mA<br>$T_C$ = +25°C | | -1.2 | V |
| High-level input current | $I_{IH1}$ | $V_{CC}$ = 5.5 V, $V_{IH}$ = 2.7 V | | 50 | µA |
| | $I_{IH2}$ | $V_{CC}$ = 5.5 V, $V_{IH}$ = 5.5 V | | 1000 | µA |
| Low-level input current | $I_{IL}$ | $V_{CC}$ = 5.5 V, $V_{IL}$ = 0.5 | | -2.0 | mA |
| Short circuit output current | $I_{OS}$ | $V_{CC}$ = 5.5 V, $V_{OS}$ = 0.0 V<br>(Not more than one output shorted at a time) | -40 | -150 | mA |
| Low-level supply current | $I_{CCL}$ | $V_{CC}$ = 5.5 V<br>$V_I$ = 0.0 V | | 75 | mA |
| Output rise time (applied to leading and trailing edge) | $T_{RO}$ | $V_{CC}$ = 5.0 V<br>$TR_I \leq 3$ ns<br>$C_L$ = 50 pF<br>$R_L$ = 500 ohms | | 4 | ns |



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Part number: The part number shall be in the following format:

| M83532 | /02 | A | 001 | A |
|---|---|---|---|---|
| Basic specification identifier | Specification sheet number (note zero) | Case styles A,B,C, or D (see appendix) | Dash number (see table I) | Product assurance level (A or B) |

QUALITY ASSURANCE PROVISIONS:

Extent of qualification: The extent of qualification shall be as specified in MIL-D-83532, for example:

Qualification and testing of M83532/02A001A and M83532/02A031A shall be sufficient to grant qualification to all dash numbers with case style A contained in this specification sheet. Similarly, for case styles B, C, and D, qualification and testing of dash numbers -001 and -031 shall be sufficient to grant qualification to all dash numbers in this specification sheet having the same case style as the components tested.

Product assurance level: Product assurance level A shall not extend to level B. Level B shall not extend to level A.

Supersession data: MIL-D-83532/2 supersedes DESC drawing numbers 84087 and 85013 when a qualified source is available.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 19, 99
DLA - ES

User activities:
Navy - AS, CG, MC, SH

Preparing activity:
Air Force - 85

Agent:
DLA - ES

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